Satyam Jay

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Education

Indian Institute of Technology, Delhi Aug 2023 — Present
PhD, Computer Science *Delhi, India*

- Research:
  - **Online Reconfiguration of Data Streaming Systems** (under review, EuroSys): on-the-fly rescaling of stream engines with minimal latency impact.
  - **Querying Spatio-Temporal Databases** (under review, SIGMOD): trajectory query engine modeling object motion as piecewise polynomials.
- Teaching Assistant: Cloud Computing, Software Design Practices, Operating Systems, Database.

Indian Institute of Technology, Delhi July 2020 — July 2022
M.Tech, Computer Science *Delhi, India*

- Major Project:- Autonomous Vehicle Simulation in Unity and CARLA
- Coursework: Adv. Data Structures, Software Systems, Functional Programming, Machine Learning, NLP, Digital Systems
- TA: Machine Learning, Intro to AI

Work

Machine Learning Engineer July 2022 — July 2023
Pepperfy *Mumbai, India*

- Enhanced recommendation engine by integrating user purchase history and search history data in the existing algorithm
- Built a prototype reverse image search engine using feature vectors from DL models

Software Engineering Internship Aug 2019 — Dec 2019
IBM *Bengaluru, India*

- Built an automated system to detect and report HIPAA-compliance violations across the compute infrastructure.

Projects

Author, Tensile github.com/satyamjay-iitd/tensile_flink

- Designed a novel protocol for online reconfiguration (on-the-fly rescaling) of stream processing with minimal latency impact, supporting job-DAG execution and fault recovery.
- Showed the protocol is engine-agnostic by implementing it in both a custom Rust engine and Apache Flink 1.17; evaluated on Nexmark against DRRS, Meces, and Megaphone with up to 20× lower reconfiguration latency.

Author, TrQL github.com/satyamjay-iitd/frameql

- Rust query engine + DSL for spatio-temporal video queries; models object trajectories as piecewise quadratic polynomials and evaluates an interval-valued algebra by root-finding, up to 126× faster than SOTA with 10× less memory.

Author, Streamary github.com/satyamjay-iitd/streamary

- Streaming SQL engine in Rust; parses SQL through an Apache Calcite / DataFusion planner into a physical plan, generates operator code, and executes it on a custom actor runtime.

Additional Projects: ReActor (Rust distributed actor framework with job manager and RPC control plane), Aperture (distributed Rust video-analytics pipeline across Jetson-edge and server; a DeepStream alternative with TensorRT / ONNX inference), and an LLVM-based C compiler.

Skills

- **Languages**: Rust, Python, Java, C/C++, SQL
- **Systems & Distributed**: Async Rust (Tokio), gRPC / tarpc, Docker, Kubernetes/K3s, Linux, OpenTelemetry
- **Data & Query Engines**: Apache Flink, DataFusion / Arrow, Apache Calcite, DuckDB, query planning & codegen
- **GPU & Video ML**: PyTorch, TensorRT, ONNX Runtime, CUDA (NVDEC/NVENC), NVIDIA DeepStream, Jetson
- **Tooling**: LLVM, Rust FFI bindings, Criterion benchmarking, Git, Typst/LaTeX